DamageTriage: Project Specification & Viva Guide

Multi-Modal SAR + Optical Building Damage Assessment & Calibrated Operational Triage

1. DATASET ARCHITECTURE & INGESTION DETAILS

The system is benchmarked on the QuickQuakeBuildings (QQB) dataset, curated from the 6 February 2023 Turkiye-Syria Earthquake sequence (Magnitude 7.8 Mw) in the Islahiye and Hatay sectors.

Specification Metric	Benchmark Value / Detail
Benchmark Dataset	QuickQuakeBuildings (QQB) - 2023 Turkiye Earthquake
Total Labeled Buildings	4,029 individual building footprints (OpenStreetMap polygons)
Ground Truth Distribution	Binary: 3,860 Intact (95.8%) vs 169 Damaged / Collapsed (4.2%)
Class Imbalance Ratio	23 : 1 (~96% Intact, ~4% Damaged) - Extreme Natural Imbalance
Geographical Folds	5 Geographically Disjoint Folds (~806 buildings each)
Held-Out Test Fold	fold-2.txt (807 buildings - strictly unseen during training)
Pre-Disaster Optical	RGB Visible Satellite Imagery (0.5m GSD resolution)
Post-Disaster Optical	RGB Visible Rubble/Collapse Imagery (0.5m GSD resolution)
Pre-Disaster SAR	Sentinel-1 C-band Radar (10m GSD, VV/VH polarization channels)
Post-Disaster SAR	Sentinel-1 C-band Radar Pass (Loss of wall double-bounce scattering)

2. END-TO-END METHODOLOGY (HOW IT WAS BUILT)

- **Phase 1: Multi-Modal Ingestion:** PyTorch QQBDataset pipeline ingesting 4 paired HDF5 (.mat) arrays per building footprint. Resampled to 128x128 pixels, converted SAR to dB scale, and enforced disjoint fold separation to prevent spatial autocorrelation leakage.
- **Phase 2: Siamese Multi-Branch Backbone:** Constructed 4-branch ResNet-18 feature extractors producing 128-d latent embeddings. Built a Learnable Gated Attention Fusion network with a softmax scalar gating mechanism: e_fused = sum(g_i * e_i).
- **Phase 3: Imbalance & Uncertainty Calibration:** Combated 96:4 imbalance via Binary Focal Loss (alpha=0.25, gamma=2.0) and Weighted Random Sampling. Implemented post-hoc Temperature Scaling fitted via L-BFGS optimization on validation Negative Log-Likelihood.
- **Phase 4: Explainability & Deployment:** Extracted decoupled Grad-CAM activation maps by hooking Layer-4 across independent sensor branches. Deployed interactive Streamlit command center and automated PDF incident dispatch reporting.

3. CORE TECHNICAL SPECIFICATIONS & TRIAGE LOGIC

Rather than naive 50% thresholding, DamageTriage establishes a 4-tier decision-theoretic operational triage queue from calibrated probabilities P(damaged):

Calibrated Probability	Operational Triage Tier & Search-and-Rescue Protocol
P(damaged) < 0.20	INTACT: Structurally sound; bypass heavy physical search.
0.20 <= P < 0.50	PROBABLY INTACT: Low structural risk; secondary sweep.
0.50 <= P < 0.80	PROBABLY DAMAGED: Moderate risk; priority drone recon.
P(damaged) >= 0.80	DAMAGED: Critical structural collapse; immediate heavy rescue dispatch.

Empirical Modality Ablation (SAR + Optical vs Optical-Only):

- **False Alarm Reduction:** In held-out test evaluations (top 20 priority queue), Optical-only classification suffered a 25.0% False Positive Rate (5 intact buildings falsely prioritized due to shadows and tarpaulins).
- **Fusion Performance:** SAR + Optical Fusion achieved a 0.0% False Positive Rate (100% precision in top priority queue) because radar verified the physical disappearance of vertical dihedral double-bounce reflections.

4. KEY SCIENTIFIC REFERENCES & RESEARCH PAPERS

- **[1] Lin et al. (2017):** 'Focal Loss for Dense Object Detection', IEEE ICCV. (Modulating factor suppressing easy negative gradients

in 96:4 imbalanced data).

- **[2] Guo et al. (2017):** 'On Calibration of Modern Neural Networks', ICML. (L-BFGS Temperature Scaling formulation to optimize Expected Calibration Error without altering ROC ranking).
- **[3] Selvaraju et al. (2017):** 'Grad-CAM: Visual Explanations from Deep Networks', IEEE ICCV. (Gradient-weighted class activation mapping adapted for Layer-4 Siamese branch hooks).
- **[4] QuickQuakeBuildings Benchmark (2023):** Multi-modal satellite building damage dataset for the 2023 Kahramanmaras, Turkiye earthquake sequence.
- **[5] Sentinel-1 SAR User Handbook:** European Space Agency (ESA). Synthetic Aperture Radar C-band microwave backscatter properties and dihedral corner reflection principles.

5. MOST PROBABLE VIVA DEFENSE QUESTIONS & ANSWERS

Q1: Why is SAR (Radar) necessary if high-resolution Optical imagery is already available?

Optical can
surface col
directly det
optical false

Q2: The QQB dataset has binary labels (0 and 1). Did you invent 4 classes?

No, ground
implemente
Temperatur

Q3: Since 96% of buildings are intact, how did you prevent the model from predicting all 0s?

We solved
frequencies
the network

Q4: How does Grad-CAM work across multiple sensor branches?

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We attach
radar rubbl

Q5: Why did you use geographically disjoint folds instead of random train-test split?

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causes spa
to unseen g